The interval minimizes negative-exponent Riesz capacity ratios

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Abstract

For a compact set $K \subset \mathbb{R}$ containing at least two points, let $C_t(K)$ be the tth root of the maximum of $\iint |x-y|^t \, d\mu(x)\, d\mu(y)$ over probability measures on K. We prove that, for every $0 < r < s < 1$, the ratio $C_s(K)/C_r(K)$ is minimized by an interval, with equality only for nondegenerate closed intervals. This resolves the interval conjecture of Clark and Laugesen, restated by Fan. The main step is a comparison of energies of monotone images of beta measures. After a logistic change of coordinates, the relevant pair distribution is a positive mixture of cells from Poisson partitions. A likelihood-ratio comparison orders the mixing rates, and refinement of the partitions gives the energy comparison by concavity. The argument also derives the beta equilibrium measure and the interval capacity formula.

1 The capacity inequality

For a Borel probability measure μ with compact support, write

$$\mathcal{E}_t(\mu) = \iint |x-y|^t \, d\mu(x)\, d\mu(y), \qquad t > 0.$$

For a nonempty compact set $K \subset \mathbb{R}$, define

$$U_t(K) = \max_{\mu \in \mathcal{P}(K)} \mathcal{E}_t(\mu), \qquad C_t(K) = U_t(K)^{1/t}.$$

The maximum exists: the probability measures on K form a weakly compact space, and the bounded continuous kernel makes the energy continuous. For the latter assertion, weak convergence of the product measures follows by uniformly approximating continuous functions on $K \times K$ by finite sums of functions $f(x)g(y)$. If K contains two distinct points, placing half the mass at each shows that $U_t(K) > 0$.

Theorem 1.1. *Let $K \subset \mathbb{R}$ be compact and contain at least two points. For every $0 < r < s < 1$,*

$$\frac{C_s(K)}{C_r(K)} \geq \frac{C_s([-1,1])}{C_r([-1,1])}, \qquad C_t([-1,1]) = \left[\frac{\Gamma((1+t)/2)\Gamma(1-t/2)}{\sqrt{\pi}}\right]^{1/t} \quad (0 < t < 1). \tag{1}$$

Equality holds if and only if K is a nondegenerate closed interval.

In the conventional Riesz notation, $C_t = \mathrm{Cap}_{-t}$. Thus the inequality is the one-dimensional negative-exponent case $p = -s$, $q = -r$ of the interval conjecture in Clark and Laugesen [1, Conjecture 12]; Fan states exactly this formulation as Conjecture 4.1 [2]. Fan's Theorem 4.3 proves positivity of equilibrium masses for finite subsets of the line, supporting numerical

tests of the conjecture. Our argument applies directly to every compact set, including infinite and disconnected sets.

We first prove a comparison for monotone maps. For $0 < t < 1$, set

$$a_t = \frac{1-t}{2}, \qquad d\beta_t(x) = \frac{[x(1-x)]^{a_t-1}}{B(a_t, a_t)}\,dx, \quad 0 < x < 1, \qquad e_t = \mathcal{E}_t(\beta_t). \tag{2}$$

Here $B(a,b) = \Gamma(a)\Gamma(b)/\Gamma(a+b)$. In particular $e_t > 0$. We will derive, rather than assume, that $e_t = U_t([0,1])$. For a bounded nondecreasing map $T : (0,1) \to \mathbb{R}$, define

$$H_t(T) = \left[\frac{\mathcal{E}_t(T_\#\beta_t)}{e_t}\right]^{1/t}. \tag{3}$$

The pushforward $T_\#\beta_t$ is the distribution of $T(X)$ when X has distribution β_t.

Theorem 1.2 (Monotone transport comparison)**.** *If $T : (0,1) \to \mathbb{R}$ is bounded and nondecreasing and $0 < r < s < 1$, then*

$$H_r(T) \le H_s(T). \tag{4}$$

Equality holds if and only if $T(x) = A + Bx$ for Lebesgue-almost every x, with $B \ge 0$.

2 Random intervals and averaging

For $0 < x < y < 1$, let

$$D_T(x,y) = \frac{T(y) - T(x)}{y - x}.$$

The ordered pair distribution

$$dP_t(x,y) = \frac{2}{e_t}(y-x)^t\,d\beta_t(x)\,d\beta_t(y), \qquad 0 < x < y < 1, \tag{5}$$

is a probability measure. The factor 2 accounts for both orders of a pair of independent beta variables. Directly from the definitions,

$$H_t(T)^t = \int D_T(x,y)^t\,dP_t(x,y). \tag{6}$$

Let

$$h(u) = \frac{1 + \tanh u}{2}, \qquad \Delta(u,z) = h(u+z) - h(u), \qquad u \in \mathbb{R},\ z > 0. \tag{7}$$

We use Poisson partitions only through the elementary properties proved below. A homogeneous Poisson process of rate $\lambda > 0$ on $\mathbb{R}$ has independent counts on disjoint intervals, and the count in an interval of length L has Poisson distribution with mean λL. Its consecutive points partition $\mathbb{R}$ into bounded cells almost surely. Applying h gives a countable partition $\mathcal{Q}_\lambda$ of $(0,1)$, apart from its cell endpoints.

Lemma 2.1 (The cell containing a uniform point)**.** *Choose V uniformly in $(0,1)$, independently of $\mathcal{Q}_\lambda$. Write the cell containing V as $(h(u), h(u+z))$. The law of (u,z) is*

$$\lambda^2 e^{-\lambda z}\Delta(u,z)\,du\,dz. \tag{8}$$

Proof. The variable $v = h^{-1}(V)$ has density $h'(v)$ on $\mathbb{R}$ and is independent of the process. Conditional on v, the distances to the nearest Poisson points on its left and right are independent exponential variables of rate λ: each survival probability is the probability of no Poisson point in the corresponding interval. Thus, for endpoints u and $u+z$ with $u < v < u+z$, their joint density is $\lambda^2 e^{-\lambda z}\,du\,dz$. Integrating over the sampled point gives

$$\lambda^2 e^{-\lambda z} \int_u^{u+z} h'(v)\,dv = \lambda^2 e^{-\lambda z}\Delta(u,z),$$

which proves (8).

For an explicit normalization check, $h' \geq 0$ and $\int_{\mathbb{R}} h' = 1$, so Tonelli's theorem gives

$$\int_{\mathbb{R}} \Delta(u,z)\,du = \int_{\mathbb{R}} \int_0^z h'(u+w)\,dw\,du = z. \tag{9}$$

Consequently the integral of (8) is $\int_0^\infty \lambda^2 z e^{-\lambda z}\,dz = 1$. □

For a bounded nondecreasing T, its one-sided limits exist, and

$$L_T = T(1-) - T(0+) \geq 0.$$

Changing T at its at most countably many discontinuities has no effect on any energy in this paper. Take a right-continuous version, and let ν be its finite Lebesgue–Stieltjes measure on $(0,1)$. Then $\nu((0,1)) = L_T$. Almost surely no Poisson endpoint is an atom of ν, since ν has at most countably many atoms and each fixed location has probability zero of being a Poisson point. For a cell $J = (x,y)$ with such endpoints, put

$$S_J = \frac{\nu(J)}{|J|} = D_T(x,y), \qquad A_{\mathcal{Q}}(v) = S_J \quad (v \in J).$$

For almost every partition, countable additivity gives

$$\int_0^1 A_{\mathcal{Q}}(v)\,dv = \sum_{J\in\mathcal{Q}} \nu(J) = L_T. \tag{10}$$

In particular, for $0 < p < 1$, Jensen's inequality yields

$$0 \leq \int_0^1 A_{\mathcal{Q}}(v)^p\,dv \leq L_T^p. \tag{11}$$

This also covers singular Stieltjes measures and maps with jumps.

Lemma 2.2 (Refinement comparison)**.** *If $0 < \lambda_1 \leq \lambda_2$, the Poisson partitions can be coupled so that $\mathcal{Q}_{\lambda_2}$ refines $\mathcal{Q}_{\lambda_1}$. For every $0 < p < 1$ and bounded nondecreasing T,*

$$\mathbb{E}\int_0^1 A_{\mathcal{Q}_{\lambda_1}}(v)^p\,dv \geq \mathbb{E}\int_0^1 A_{\mathcal{Q}_{\lambda_2}}(v)^p\,dv. \tag{12}$$

Proof. Superpose a rate-λ_1 process and an independent rate-$(\lambda_2 - \lambda_1)$ process, omitting the second process when the rates are equal. Their union is a rate-λ_2 process by the addition rule for independent Poisson counts. Every coarse cell is bounded in the u coordinate and hence contains only finitely many added points almost surely. If it splits into fine cells $J_1, \ldots, J_m$, its slope is

$$S_J = \sum_{i=1}^m \frac{|J_i|}{|J|} S_{J_i}.$$

Concavity of z^p gives $|J|S_J^p \geq \sum_i |J_i| S_{J_i}^p$. Summing over the countably many coarse cells, then taking expectations, proves (12). The sums are nonnegative and bounded by (11). □

3 The beta pair distribution as a Poisson mixture

Lemma 3.1 (Mixture representation)**.** *Under the change of variables $x = h(u)$, $y = h(u+z)$, the probability measure P_t in* (5) *has density*

$$\frac{\Delta(u,z)(\sinh z)^{t-1}}{Z_t}\,du\,dz, \qquad Z_t = \int_0^\infty z(\sinh z)^{t-1}\,dz. \tag{13}$$

Let $\alpha = 1-t$ and

$$b_n(\alpha) = \frac{(\alpha)_n}{n!}, \quad \lambda_n(\alpha) = 2n+\alpha, \quad W(\alpha) = \sum_{n=0}^{\infty}\frac{b_n(\alpha)}{\lambda_n(\alpha)^2}, \quad w_n(\alpha) = \frac{b_n(\alpha)}{\lambda_n(\alpha)^2 W(\alpha)}. \tag{14}$$

Here $(\alpha)_0 = 1$ and $(\alpha)_n = \alpha(\alpha+1)\cdots(\alpha+n-1)$. Then P_t is the mixture of the cell laws (8)*, with rate $\lambda_n(\alpha)$ chosen with probability $w_n(\alpha)$.*

Proof. Use

$$h'(u) = \frac{1}{2\cosh^2 u}, \qquad h(u)(1-h(u)) = \frac{1}{4\cosh^2 u}, \qquad \Delta(u,z) = \frac{\sinh z}{2\cosh u\cosh(u+z)}.$$

The unnormalized ordered-pair density transforms according to

$$\begin{aligned}(y-x)^t[x(1-x)y(1-y)]^{-(1+t)/2}\,dx\,dy \\ = 2^{t+1}(\sinh z)^{t-1}\Delta(u,z)\,du\,dz.\end{aligned}$$

Equation (9) supplies the normalization in (13). The integral Z_t is finite: its integrand is of order z^t at zero and decays exponentially times z at infinity.

For every $z > 0$, the binomial series gives the positive expansion

$$(\sinh z)^{-\alpha} = 2^\alpha e^{-\alpha z}(1-e^{-2z})^{-\alpha} = 2^\alpha\sum_{n=0}^{\infty} b_n(\alpha)e^{-\lambda_n(\alpha)z}. \tag{15}$$

Tonelli's theorem and (9) imply $Z_t = 2^\alpha W(\alpha)$, in particular $0 < W(\alpha) < \infty$. Substitution into (13) expresses that density as

$$\sum_{n=0}^{\infty} w_n(\alpha)\,\lambda_n(\alpha)^2 e^{-\lambda_n(\alpha)z}\Delta(u,z).$$

Lemma 2.1 identifies each summand's normalized density. □

Lemma 3.2 (Ordering the mixing rates)**.** *If $0 < \beta < \alpha < 1$, random integers with distributions $w_n(\beta)$ and $w_n(\alpha)$ can be coupled as $N_\beta \le N_\alpha$. In this coupling,*

$$2N_\beta + \beta < 2N_\alpha + \alpha.$$

Proof. Apart from a positive factor independent of n, the likelihood ratio is

$$\frac{w_n(\beta)}{w_n(\alpha)} = \frac{W(\alpha)}{W(\beta)}\frac{(\beta)_n}{(\alpha)_n}\left(\frac{2n+\alpha}{2n+\beta}\right)^2. \tag{16}$$

The Pochhammer ratio decreases with n, since the ratio of successive terms is $(n+\beta)/(n+\alpha) < 1$. The last factor also decreases, as $(2n+\alpha)/(2n+\beta) = 1 + (\alpha-\beta)/(2n+\beta)$. Thus the likelihood ratio decreases.

For completeness, put $q_n = w_n(\beta)$ and $p_n = w_n(\alpha)$. For each integer $k \geq 0$, $q_i p_j \geq p_i q_j$ whenever $i \leq k < j$. Summing shows

$$\left(\sum_{i\leq k} q_i\right)\left(1 - \sum_{i\leq k} p_i\right) \geq \left(\sum_{i\leq k} p_i\right)\left(1 - \sum_{i\leq k} q_i\right),$$

and hence $\sum_{i\leq k} q_i \geq \sum_{i\leq k} p_i$. Applying the two discrete inverse distribution functions to the same uniform random number gives $N_\beta \leq N_\alpha$. The strict rate inequality follows from $\beta < \alpha$. □

Proposition 3.3 (Concave moments of chord slopes)**.** *For $0 < r < s < 1$, every bounded nondecreasing T, and every $0 < p < 1$,*

$$\int D_T^p \, dP_s \geq \int D_T^p \, dP_r. \tag{17}$$

Also, for every $0 < t < 1$,

$$\int D_T \, dP_t = L_T, \qquad \int D_T^p \, dP_t \leq L_T^p \quad (0 < p < 1). \tag{18}$$

Proof. Take $\alpha = 1 - r$ and $\beta = 1 - s$ in Lemma 3.2. Conditional on the coupled rates, use Lemma 2.2 to couple the smaller-rate partition as a coarsening of the larger-rate partition. Choose the same independent uniform point V to select the cells. Lemma 3.1 identifies their endpoint laws as P_s and P_r, respectively. Conditional Jensen, as in (12), proves (17) after averaging over the rates. Equations (10) and (11), averaged over the mixture, prove (18). □

Proof of Theorem 1.2. By the power-mean inequality on the probability space P_s, followed by Proposition 3.3 with $p = r$,

$$\left(\int D_T^s \, dP_s\right)^{1/s} \geq \left(\int D_T^r \, dP_s\right)^{1/r} \geq \left(\int D_T^r \, dP_r\right)^{1/r}. \tag{19}$$

All moments are finite by (18). Together with (6), this proves the inequality.

If equality holds, equality in the first step of (19) implies that D_T is constant P_s-almost surely; this follows also from strict concavity of $z^{r/s}$ applied to D_T^s. The density of P_s is strictly positive on $0 < x < y < 1$. It follows that $T(y) - T(x) = B(y - x)$ for Lebesgue-almost every ordered pair, for a constant $B \geq 0$. By symmetry this holds for almost every pair in $(0, 1)^2$. Fubini's theorem then implies that $T(x) - Bx$ is almost everywhere constant. Conversely, if $T(x) = A + Bx$ almost everywhere, then $D_T = B$ almost everywhere, and $H_t(T) = B$ for every t. This includes constant maps. □

4 Equilibrium of the interval and arbitrary compact sets

Lemma 4.1 (Monotone realization of a measure)**.** *Let μ be a probability measure supported on a compact subset of $\mathbb{R}$, and let $0 < t < 1$. There is a bounded nondecreasing Borel map $T : (0, 1) \to \mathbb{R}$ such that $T_\# \beta_t = \mu$. For every $0 < u < 1$, $T_\# \beta_u$ is supported on* $\operatorname{supp} \mu$.

Proof. Let F_t be the continuous strictly increasing distribution function of β_t. It maps a beta-distributed variable to a uniform variable. If $Q_\mu(v) = \inf\{x : \mu((-\infty, x]) \geq v\}$ is the quantile function, take $T = Q_\mu \circ F_t$. The usual inverse-distribution identity gives $T_\# \beta_t = \mu$. In particular, $T(x) \in \operatorname{supp} \mu$ for β_t-almost every x. All the measures β_u, $0 < u < 1$, have strictly positive densities on $(0, 1)$ and thus have the same null sets. The support assertion follows. □

Proposition 4.2 (Interval equilibrium and its energy)**.** *For $0 < t < 1$, β_t maximizes $\mathcal{E}_t$ among all probability measures on $[0,1]$, and*

$$e_t = U_t([0,1]) = \frac{\Gamma((1+t)/2)\Gamma(1-t/2)}{2^t\sqrt{\pi}}. \tag{20}$$

Its potential $\int_0^1 |x-y|^t\, d\beta_t(y)$ equals e_t at every $x \in [0,1]$.

Proof. For any probability measure μ on $[0,1]$, Lemma 4.1 gives a monotone map T taking values in $[0,1]$ almost everywhere. Thus $L_T \le 1$, and (6) and (18) give

$$\frac{\mathcal{E}_t(\mu)}{e_t} = \int D_T^t\, dP_t \le L_T^t \le 1.$$

This proves the equilibrium assertion without using a preexisting interval-capacity formula.

Let $V_t(x) = \int |x-y|^t\, d\beta_t(y)$. For $x \in [0,1]$, maximality applied to $(1-\varepsilon)\beta_t + \varepsilon\delta_x$ gives

$$0 \ge \frac{d}{d\varepsilon}\mathcal{E}_t\big((1-\varepsilon)\beta_t + \varepsilon\delta_x\big)\Big|_{\varepsilon=0+} = 2(V_t(x) - e_t).$$

The function V_t is continuous, β_t has full support on $[0,1]$, and $\int V_t\, d\beta_t = e_t$. Therefore $V_t(x) = e_t$ everywhere on $[0,1]$. At zero this gives, with $a = (1-t)/2$,

$$e_t = \frac{B(a+t,a)}{B(a,a)} = \frac{\Gamma((1+t)/2)\Gamma(1-t)}{\Gamma((1-t)/2)}.$$

The gamma duplication formula yields (20). □

Proof of Theorem 1.1. Fix $0 < r < s < 1$ and choose an r-energy maximizing measure μ on K. By Lemma 4.1, choose T with $T_\#\beta_r = \mu$; the measure $T_\#\beta_s$ is also supported on K. The transport comparison and Proposition 4.2 give

$$\begin{aligned}\frac{C_r(K)}{C_r([0,1])} &= \left[\frac{\mathcal{E}_r(T_\#\beta_r)}{e_r}\right]^{1/r} \\ &\le \left[\frac{\mathcal{E}_s(T_\#\beta_s)}{e_s}\right]^{1/s} \le \frac{C_s(K)}{C_s([0,1])}.\end{aligned}$$

The capacities in the denominators and $C_r(K)$ are positive, so this is the desired ratio inequality. Translating a set does not change its energy, and scaling it by $b > 0$ multiplies C_t by b. Thus $[0,1]$ and $[-1,1]$ have the same ratio; scaling (20) also gives the gamma formula in (1).

For equality, both steps of the displayed comparison must be equalities. Theorem 1.2 gives $T(x) = A + Bx$ almost everywhere. Here $B > 0$, because $U_r(K) > 0$. Thus μ is the affine image of β_r, and its support is the interval $J = [A, A+B] \subset K$. Its potential equals $U_r(J) = U_r(K)$ on J by Proposition 4.2 and scaling. If K contained a point z outside J, choose the endpoint $b = A + B$ when $z > A + B$, and the endpoint $b = A$ when $z < A$. For every $y \in J$, we have $|z-y| > |b-y|$. Hence the potential at z would strictly exceed the potential at b, namely $U_r(K)$. Adding a sufficiently small positive mass at z, as in the preceding variation calculation, would increase the energy, a contradiction. Consequently $K = J$. Conversely, every nondegenerate closed interval attains equality by translation and scaling. □

5 Examples, attribution, and verification

For two points at distance $d > 0$, weights a and $1 - a$ give energy $2a(1 - a)d^t$, maximized at $a = 1/2$. Therefore

$$C_t(\{x, y\}) = d\, 2^{-1/t}, \qquad \frac{C_s(\{x, y\})}{C_r(\{x, y\})} = 2^{1/r-1/s}.$$

Theorem 1.1 makes this strictly greater than the interval ratio. More generally the inequality is strict for every finite set with at least two points, every disconnected compact set, and every compact set that is not an interval. For an affine map $T(x) = A + Bx$ with $B \geq 0$, the transport quantities satisfy $H_t(T) = B$, illustrating equality in the stronger transport theorem.

The conjecture and the classical interval benchmark are attributed to Clark and Laugesen [1] and Fan [2]. The contribution here is the monotone transport comparison, its Poisson-partition proof, and its application to the full interval conjecture with equality classification. The proof uses standard beta integrals, Poisson superposition, likelihood-ratio ordering, and Jensen's inequality; the required identities and comparison arguments are included above. The accompanying literature review records the sources and public repositories checked for earlier resolutions.

The proof is analytic and requires no numerical certificate. Optional verification code checks the mixture normalization and finite quantile-transport examples. The manuscript and its independent internal mathematical checks were produced by GPT-6 Astra. The package contains the LaTeX source and the proof audit; it makes no claim of proof-assistant formalization or external peer review.

References

[1] C. Clark and R. S. Laugesen, *Maximizing Riesz capacity ratios: conjectures and theorems*, SIAM Journal on Mathematical Analysis **57** (2025), 5683–5717. doi:10.1137/24M171992X. Preprint: arXiv:2410.14809.

[2] Q. Fan, *Riesz capacity ratios with negative exponents*, arXiv:2609.11186v1 (2026).